

Supervised Neural Network Project: GNN-Based BERT for Understanding Context from Music

Course: Neural Networks (CSE425 / EEE474 / CSE715)

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Submission Deadline: **2nd October, 2026**

1 Project Motivation

Music is a **multi-layered signal** where “context” spans melody, harmony, rhythm, lyrics, metadata tags, and listener-described semantics. A single clip may simultaneously express:

- **Genre and era** (e.g., Jazz, 1960s)
- **Mood and emotion** (e.g., melancholic, high arousal)
- **Harmonic structure** (chord progressions, key changes)
- **Lyrical semantics** (themes, sentiment in vocals)
- **Timbral texture** (instrumentation, production style)

Pure sequence models (CNN/RNN on spectrograms) capture local patterns but often miss **relational structure**: how chord $C \rightarrow G \rightarrow Am$ relates to a lyrical theme, or how repeated segments form a song graph.

Goal: Build a hybrid **BERT + Graph Neural Network (GNN)** system that understands musical context by combining:

1. **BERT** — contextual language representations from lyrics, tags, and natural-language music descriptions (MusicCaps).
2. **GNN** — message passing on music structure graphs (chord transitions, segment similarity, co-occurrence of audio events).

Unlike generative music projects, this assignment focuses on **understanding and prediction**: multi-label tagging, emotion regression, and cross-modal alignment between audio structure and text.

2 Problem Definition

Let a music track be represented as a tuple:

$$\mathcal{T} = (\mathbf{X}_{\text{audio}}, \mathbf{X}_{\text{text}}, \mathbf{G}, \mathbf{y}),$$

where:

- $\mathbf{X}_{\text{audio}}$ = log-mel spectrogram or chroma features over T frames,
- $\mathbf{X}_{\text{text}}$ = tokenized lyrics, user tags, or caption tokens,

- $\mathbf{G} = (\mathcal{V}, \mathcal{E})$ = music structure graph (nodes = segments/chords/events; edges = transitions or similarity),
- $\mathbf{y}$ = context labels (genre, mood tags, valence/arousal, etc.).

BERT text encoder (frozen or fine-tuned) maps text to contextual embeddings:

$$\mathbf{H}_{\text{text}} = \text{BERT}(\mathbf{X}_{\text{text}}) \in \mathbb{R}^{L \times d}.$$

GNN encoder with L layers updates node features via message passing:

$$\mathbf{h}_i^{(l+1)} = \sigma\left(\mathbf{W}^{(l)} \cdot \text{AGG}(\mathbf{h}_i^{(l)}, \{\mathbf{h}_j^{(l)} : j \in \mathcal{N}(i)\})\right),$$

where $\mathbf{h}_i^{(0)}$ is initialized from audio segment embeddings or chord features.

Fusion readout combines graph-level representation $\mathbf{g}$ and text CLS vector $\mathbf{t}$:

$$\mathbf{z} = \text{Fusion}(\mathbf{g}, \mathbf{t}), \quad \hat{\mathbf{y}} = \sigma(\mathbf{W}\mathbf{z} + \mathbf{b}).$$

Objective (multi-label context understanding):

$$\mathcal{L} = - \sum_{k=1}^K \left[y_k \log \hat{y}_k + (1 - y_k) \log(1 - \hat{y}_k) \right] + \lambda \mathcal{L}_{\text{aux}}.$$

3 Dataset Requirements

Students must use **at least one primary audio dataset** and **one text/tag dataset** from Table 1. Advanced tasks should combine **FMA or MagnaTagATune** with **MusicCaps** or **DEAM**.

Recommended primary pairing

- **Medium/Hard:** FMA-medium (audio + genre/tags) + chord/segment graphs built from chroma features.
- **Advanced:** MusicCaps captions (BERT) aligned with audio graphs from corresponding 10s clips.
- **Emotion extension:** DEAM for valence/arousal regression as auxiliary $\mathcal{L}_{\text{aux}}$.

Preprocessing pipeline

1. **Audio:** Resample to 22,050 Hz; extract log-mel spectrogram (128 bins) or chroma (12 bins); normalize per track.
2. **Segmentation:** Split each track into fixed windows (e.g., 5–10 s) or beat-synchronous segments using librosa.
3. **Graph construction:**
 - *Chord-transition graph:* nodes = unique chords; edges = observed transitions weighted by count.

Table 1: Recommended datasets for GNN-BERT music context understanding.

Dataset Name	Context / Label Coverage	Task	Official Link
FMA (small / medium)	Genre (8–16 classes), top tags, artist & album metadata	1–3	FMA Download
MagnaTagATune	188 multi-label tags (genre, mood, instrument); 25,877 clips	1, 3	MagnaTagATune
GTZAN	10 genres; 1,000 tracks (30 s clips); standard Easy baseline	1–2	GTZAN Download
DEAM	Continuous valence & arousal (1–9) every 0.5 s	3–4	DEAM Download
MusicCaps	5,521 clips with expert natural-language captions (Google)	1, 4	MusicCaps
Million Song Dataset + tagtraum	Last.fm-style tags for 188,916 tracks (Echo Nest audio features)	1, 3	MSD Download
Lakh MIDI Clean	Symbolic MIDI for chord-transition & melody graph construction	2–3	LMD Download
EmoMusic	1,000 clips with quadrant emotion labels (happy/sad × calm/energetic)	3	EmoMusic

Task key: 1 = BERT tags — 2 = GNN structure — 3 = GNN-BERT fusion — 4 = contrastive retrieval. Use ≥ 1 audio dataset and ≥ 1 text/tag source.

- *Segment graph:* nodes = time segments; edges = temporal adjacency + cosine similarity of MFCC/chroma $> \tau$.
4. **Text:** Tokenize lyrics/tags/captions with BERT tokenizer (max length 128–256); pad/truncate.
 5. **Splits:** Use official FMA / MagnaTagATune splits; for DEAM use standard train/val partition; **no artist leakage** across train/test when possible.

4 Model Tasks and Mathematical Formulation

4.1 Task 1 (Easy): BERT Baseline for Music Tag Understanding

Goal. Implement a **BERT-based multi-label classifier** on textual music context (tags, captions, or lyrics) without graph structure.

Model.

$$\mathbf{t} = \text{BERT}_{\text{CLS}}(\mathbf{X}_{\text{text}}), \quad \hat{y}_k = \sigma(\mathbf{w}_k^\top \mathbf{t} + b_k).$$

Loss (binary cross-entropy per tag k):

$$\mathcal{L}_{\text{BERT}} = -\frac{1}{K} \sum_{k=1}^K [y_k \log \hat{y}_k + (1 - y_k) \log(1 - \hat{y}_k)].$$

Deliverables.

- BERT fine-tuning code (HuggingFace `bert-base-uncased` or `distilbert-base-uncased`)
- Results on MagnaTagATune tag subset (top-50 tags) or MusicCaps caption $\rightarrow$ tag proxy task
- Macro-F1 / Micro-F1 curves vs. training epochs
- 5 example predictions with attention visualization (optional)

4.2 Task 2 (Medium): GNN on Music Structure Graphs

Goal. Build a **GraphSAGE** or **GAT** encoder on segment/chord graphs using **audio-only** node features; predict genre or top tags.

GraphSAGE update:

$$\mathbf{h}_i^{(l+1)} = \sigma\left(\mathbf{W}^{(l)} \cdot \text{CONCAT}(\mathbf{h}_i^{(l)}, \text{MEAN}_{j \in \mathcal{N}(i)} \mathbf{h}_j^{(l)})\right).$$

Graph readout (mean pooling):

$$\mathbf{g} = \frac{1}{|\mathcal{V}|} \sum_{i \in \mathcal{V}} \mathbf{h}_i^{(L)}, \quad \hat{\mathbf{y}} = \sigma(\mathbf{W}\mathbf{g} + \mathbf{b}).$$

Deliverables.

- Graph construction scripts (chroma/MFCC segment graphs)
- GNN implementation (PyTorch Geometric)
- Genre classification on GTZAN or FMA-small
- Comparison vs. CNN baseline on mel-spectrogram

4.3 Task 3 (Hard): GNN–BERT Fusion for Multi-Context Understanding

Goal. Fuse **structural** (GNN) and **semantic text** (BERT) representations to predict **multi-label context**: genre + mood tags + (optional) valence/arousal.

Cross-attention fusion (recommended):

$$\mathbf{A} = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d}}\right), \quad \mathbf{Q} = \mathbf{g}\mathbf{W}_Q, \quad \mathbf{K} = \mathbf{H}_{\text{text}}\mathbf{W}_K,$$

$$\mathbf{z} = \text{CONCAT}(\mathbf{g}, \mathbf{A}\mathbf{H}_{\text{text}}), \quad \hat{\mathbf{y}} = \sigma(\mathbf{W}\mathbf{z}).$$

Multi-task loss:

$$\mathcal{L} = \mathcal{L}_{\text{tags}} + \alpha \|\mathbf{v} - \hat{\mathbf{v}}\|_2^2 + \beta \|\mathbf{a} - \hat{\mathbf{a}}\|_2^2,$$

where $(\mathbf{v}, \mathbf{a})$ are DEAM valence/arousal targets when available.

Deliverables.

- End-to-end GNN–BERT fusion model
- Ablation: BERT-only, GNN-only, early concat, cross-attention
- Results on FMA-medium or MagnaTagATune (Macro-F1, AUC-PR)
- t-SNE of $\mathbf{z}$ coloured by genre and mood
- 3 case studies showing graph paths + caption/lyric alignment

4.4 Task 4 (Advanced): Cross-Modal MusicCaps Alignment

Goal. Learn a **shared embedding space** between audio graphs and natural-language descriptions (MusicCaps) using contrastive learning.

InfoNCE contrastive loss for paired (graph, caption) $(\mathbf{g}_i, \mathbf{t}_i)$:

$$\mathcal{L}_{\text{NCE}} = -\log \frac{\exp(\text{sim}(\mathbf{g}_i, \mathbf{t}_i)/\tau)}{\sum_{j=1}^N \exp(\text{sim}(\mathbf{g}_i, \mathbf{t}_j)/\tau)},$$

where $\text{sim}(\mathbf{u}, \mathbf{v}) = \mathbf{u}^\top \mathbf{v} / \|\mathbf{u}\| \|\mathbf{v}\|$ and τ is temperature.

Retrieval metrics: Caption $\rightarrow$ Audio R@1, R@5, R@10; Audio $\rightarrow$ Caption R@K.

Deliverables.

- Dual-encoder GNN-BERT with contrastive training
- Retrieval evaluation table on MusicCaps test split
- 10 qualitative retrieval examples (query caption $\rightarrow$ top-3 matched clips)
- Zero-shot tag prediction from captions vs. Task 3 supervised model

5 Task Roadmap Summary

Table 2: Four-task roadmap with objectives and deliverables.

Task	Model	Key equations	Deliverables
1 (Easy)	BERT tag classifier	$\hat{y}_k = \sigma(\mathbf{w}_k^\top \mathbf{t}); \mathcal{L}_{\text{BERT}}$	BERT code; tag F1; 5 predictions
2 (Medium)	GNN on segment/chord graph	GraphSAGE update; mean readout $\mathbf{g}$	Graph builder; GNN code; vs. CNN
3 (Hard)	GNN-BERT fusion	Cross-attention; multi-task $\mathcal{L}$	Ablations; t-SNE; case studies
4 (Advanced)	Contrastive dual-encoder	InfoNCE $\mathcal{L}_{\text{NCE}}$; R@K	Retrieval table; 10 qual. ex-amples

6 Evaluation Metrics

Tag / genre classification.

$$\text{Prec}_k = \frac{TP_k}{TP_k + FP_k}, \quad \text{Rec}_k = \frac{TP_k}{TP_k + FN_k}, \quad \text{F1}_k = \frac{2 \text{Prec}_k \text{Rec}_k}{\text{Prec}_k + \text{Rec}_k}.$$

Macro-F1 = $\frac{1}{K} \sum_k \text{F1}_k$; **Micro-F1** pools globally.

AUC-PR. Area under precision-recall curve per tag; report mean AUC-PR over tags.

Emotion regression (DEAM).

$$\text{MAE}_v = \frac{1}{N} \sum_{i=1}^N |v_i - \hat{v}_i|, \quad R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}.$$

Graph coherence score (optional analysis). Measure whether high-attention edges align with repeated chord patterns:

$$S_{\text{graph}} = \frac{1}{|\mathcal{E}|} \sum_{(i,j) \in \mathcal{E}} \mathbb{I}[\cos(\mathbf{h}_i, \mathbf{h}_j) > \tau].$$

Human evaluation (Task 4). Minimum 5 listeners rate whether retrieved clip matches caption on scale $[1, 5]$.

7 Algorithms for the Tasks

Algorithm 1 Task 1: BERT Multi-Label Tag Classifier

Require: Tag corpus $\mathcal{D} = \{(\mathbf{X}_{\text{text}}^{(i)}, \mathbf{y}^{(i)})\}$
Require: Pretrained BERT, learning rate η , epochs E

- 1: Add classification head $\mathbf{W}, \mathbf{b}$ on top of CLS token
- 2: **for** epoch = 1 to E **do**
- 3: **for** each mini-batch in $\mathcal{D}$ **do**
- 4: $\mathbf{t} \leftarrow \text{BERT}_{\text{CLS}}(\mathbf{X}_{\text{text}})$
- 5: $\hat{\mathbf{y}} \leftarrow \sigma(\mathbf{W}\mathbf{t} + \mathbf{b})$
- 6: $\mathcal{L} \leftarrow \text{BCE}(\mathbf{y}, \hat{\mathbf{y}})$
- 7: Update BERT (optional) and head via $\eta \nabla \mathcal{L}$
- 8: **end for**
- 9: **end for**
- 10: **return** fine-tuned BERT tag model

Algorithm 2 Task 2: GNN Encoder on Music Segment Graph

Require: Audio tracks $\{\mathcal{T}_i\}$, label set $\mathbf{y}$

- 1: **for** each track $\mathcal{T}$ **do**
- 2: Extract segment features $\mathbf{h}_i^{(0)}$ from mel/chroma windows
- 3: Build graph $\mathbf{G} = (\mathcal{V}, \mathcal{E})$ (temporal + similarity edges)
- 4: **end for**
- 5: **for** epoch = 1 to E **do**
- 6: **for** each graph $\mathbf{G}$ in batch **do**
- 7: **for** layer $l = 0$ to $L - 1$ **do**
- 8: $\mathbf{h}_i^{(l+1)} \leftarrow \text{GraphSAGE}(\mathbf{h}_i^{(l)}, \mathcal{N}(i))$
- 9: **end for**
- 10: $\mathbf{g} \leftarrow \text{MEANPOOL}(\{\mathbf{h}_i^{(L)}\})$
- 11: $\mathcal{L} \leftarrow \text{BCE}(\mathbf{y}, \sigma(\mathbf{W}\mathbf{g}))$
- 12: Update GNN parameters
- 13: **end for**
- 14: **end for**

8 Baseline Models for Comparison

Students must compare against **at least two baselines**:

- **B1:** Majority-class / random tag predictor

Algorithm 3 Task 3: GNN–BERT Fusion for Context Understanding

Require: Paired data $(\mathbf{G}, \mathbf{X}_{\text{text}}, \mathbf{y})$

- 1: $\mathbf{H}_{\text{text}} \leftarrow \text{BERT}(\mathbf{X}_{\text{text}})$, $\mathbf{t} \leftarrow \mathbf{H}_{\text{text}}[\text{CLS}]$
 - 2: $\mathbf{g} \leftarrow \text{GNN_Readout}(\mathbf{G})$
 - 3: $\mathbf{z} \leftarrow \text{CrossAttention}(\mathbf{g}, \mathbf{H}_{\text{text}})$ $\triangleright$ or $\text{CONCAT}(\mathbf{g}, \mathbf{t})$
 - 4: $\hat{\mathbf{y}} \leftarrow \sigma(\mathbf{W}\mathbf{z} + \mathbf{b})$
 - 5: $\mathcal{L} \leftarrow \mathcal{L}_{\text{tags}} + \alpha \mathcal{L}_{\text{emotion}}$
 - 6: Backprop through fusion, GNN, and (partial) BERT
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Algorithm 4 Task 4: Contrastive GNN–BERT (MusicCaps)

Require: Paired $(\mathbf{G}_i, \text{caption}_i)$, temperature τ , batch size N

- 1: $\mathbf{g}_i \leftarrow \text{Normalize}(\text{GNN}(\mathbf{G}_i))$
 - 2: $\mathbf{t}_i \leftarrow \text{Normalize}(\text{BERT}_{\text{CLS}}(\text{caption}_i))$
 - 3: **for** each training step **do**
 - 4: Compute similarity matrix $\mathbf{S}_{ij} = \mathbf{g}_i^\top \mathbf{t}_j / \tau$
 - 5: $\mathcal{L}_{\text{NCE}} \leftarrow -\frac{1}{N} \sum_i \log \frac{\exp(\mathbf{S}_{ii})}{\sum_j \exp(\mathbf{S}_{ij})}$
 - 6: Update encoders to minimize $\mathcal{L}_{\text{NCE}}$
 - 7: **end for**
 - 8: Evaluate retrieval R@1, R@5, R@10 on held-out pairs
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- **B2:** CNN on mel-spectrogram (no graph, no text)
- **B3:** BERT-only (Task 1)
- **B4 (optional):** PCA + MLP on hand-crafted audio features

Table 3: Illustrative performance comparison (replace with your experimental results).

Model	Macro-F1	AUC-PR	MAE (emotion)	R@5 (retrieval)
Random tags	0.05	0.12	–	0.02
CNN mel-spec	0.41	0.38	1.25	–
Task 1: BERT-only	0.48	0.44	–	–
Task 2: GNN-only	0.52	0.47	1.10	–
Task 3: GNN–BERT	0.61	0.55	0.92	–
Task 4: Contrastive	0.55	0.50	–	0.38

9 Judging Rubric

Marks distribution (100 total)

Component	Marks	Notes
Task 1 (Easy)	18	BERT tags + F1 report
Task 2 (Medium)	22	GNN graph + CNN baseline
Task 3 (Hard)	22	Fusion + ablations
Task 4 (Advanced)	18	Contrastive retrieval (optional $\rightarrow$ bonus)
Report + GitHub	20	Reproducibility + paper quality
Total	100	

Table 4: Final grading rubric (maps to 15% course grade; total 100 marks).

Category	Criteria	Weight
Dataset & preprocessing	Correct splits, no leakage, graph construction documented	15%
Model implementation	BERT + GNN correct, stable training, reproducible code	25%
Context understanding quality	Strong tag/emotion/retrieval results vs. baselines	20%
Baseline comparison	≥ 2 baselines with fair experimental setup	15%
Metrics & analysis	Macro-F1, AUC-PR, ablations, visualizations	15%
Report & presentation	Clear NeurIPS/IEEE-style report, diagrams, discussion	10%

10 Final Submission Requirements

Each group must submit:

1. **GitHub repository** or ZIP with full source code
2. **Preprocessed graph samples** (at least 20 example `.pt` / `.json` graphs)
3. **Evaluation tables + plots** (F1, AUC-PR, t-SNE, retrieval examples)
4. **Final report PDF** (6–10 pages; NeurIPS / IEEE / ICML Overleaf template)
5. **Demo notebook** (`notebooks/demo_context.ipynb`) with one end-to-end inference example

GitHub project structure

```

gnn-bert-music-context/
  README.md
  requirements.txt
  config.yaml
  data/
    raw/           # FMA, MagnaTagATune, MusicCaps downloads
    processed/     # graphs, mel-spec, BERT caches
    splits/        # train/val/test JSON
  notebooks/
    eda.ipynb
    demo_context.ipynb
  src/
    audio_features.py  # mel, chroma, segmentation
    graph_builder.py   # chord + segment graphs
    bert_encoder.py
    gnn_model.py       # GraphSAGE / GAT
    fusion_model.py    # cross-attention GNN-BERT
    contrastive.py     # Task 4 InfoNCE
    train.py
    evaluate.py
  results/
    metrics.json

```



```
plots/  
  retrieval_examples/  
report/  
  final_report.pdf
```

Report template links (Overleaf)

- NeurIPS 2024: <https://www.overleaf.com/latex/templates/neurips-2024/tpsbbbrdqcmsh>
- IEEE Conference: <https://www.overleaf.com/latex/templates/ieee-conference-template>
- ICML 2025: <https://www.overleaf.com/latex/templates/icml2025-template>